MIL-R-11809B(EL)
11 August 1972
SUPERSEDING
MIL-R-11809A(EL)
27 March 1968

MILITARY SPECIFICATION

RADAR INTERROGATOR SET AN/TPX-19, 20, 26 and 27 PACKAGING AND PACKING OF

1. SCOPE

1.1 This specification covers the requirements for preservation, packaging, packing and marking of Radar Interrogator Set AN/TPX-19, 20, 26 and 27 for shipment and storage.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
T-T-871	Twine, Cotton, Wrapping
PPP-B-636	Box, Fiberboard
PPP-C-843	Cushioning Material, Cellulosic
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing)
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

FSC 5895

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MIL-E-75	Electron Tubes, Packaging Of
MIL-P-116	Preservation, Methods Of
MIL-B-117	Bags, Sleeves and Tubing, - Interior Packaging
MIL-S-19491	Semiconductor Devices, Packaging Of
MIL-B-43014	Boxes: Water-Resistant Paperboard; Folding, Set-Up, and Metal-Stayed
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging Of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards, required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein.

3.2 Dimensional data. Dimensions of packaging materials in the Bill of Material Table are based on the dimensions of the equipment cited in the Tables. When the dimensions of the equipment vary from those cited, the dimensions of the packaging materials shall be adjusted accordingly.

3.3 Preservation and packaging. Preservation and packaging shall be level A or C, as specified in the contract or order (see 6.1).

3.3.1 Level A.

3.3.1.1 Cleaning. Each Radar Interrogator Set AN/TPX-19, 20, 26 or 27 shall be cleaned in accordance with process C-1 of MIL-P-116.

3.3.1.2 Drying. Each Radar Interrogator Set AN/TPX-19, 20, 26 or 27 shall be dried in accordance with the applicable procedure of MIL-P-116.

3.3.1.3 Preservation application. None required.

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3.3.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

3.3.1.5 Technical literature and bulletins. Each technical literature and bulletin shall be individually packaged method IC-1.

3.3.1.6 Radar Interrogator Set AN/TPX-19, 20, 26 and 27. Each interrogator set shall be packaged as follows:

3.3.1.6.1 Case, Receiver-Transmitter CY-944/TPX or CY-944A/TPX with contents. Each case and contents shall be packaged in accordance with items 1 through 3 of table I and figure 1.

3.3.1.6.2 Case, Standardized Components, Electrical CY-951( )/TPX with contents. Each case and contents shall be packaged in accordance with items 1 through 3 of table II and figure 2.

3.3.1.6.3 Cable Assembly CX-2287( )/U and CX-4443/U on reel. Each cable assembly or assemblies on reel shall be packaged in accordance with items 1 and 2 of table III and figure 3.

3.3.1.6.4 Case, Antenna CY-1230( )/GPA-8 with contents. Each case and contents shall be packaged in accordance with items 3 through 6 of table III and figure 3.

3.3.1.6.5 Cable Assemblies CX-4444( )/U for use with AN/TPX-26 only. The two cable assemblies CX-4444( )/U, one 155 feet in length and the other 18 feet in length with connectors inserted in bags fabricated of material conforming to L-P-378, minimum 4 mils thick, shall be wound on a reel. The bags shall be secured to the cables with tape conforming to PPP-T-76. The cable on the reel shall be overwrapped, full width, with one thickness of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, and of sufficient length to overlap itself by a minimum of 6 inches. The overwrap shall be secured by one band of tape conforming to PPP-T-97, type IV.

3.3.1.6.6 Accessories, spares and ancillary items. Accessories, spares and ancillary items shall be packaged in accordance with items 1 through 4 of table IV.

3.3.2 Level C. Each Radar Interrogator Set AN/TPX-19, 20, 26 or 27 shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

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3.4 Packing. Packing shall be level A, B or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.1).

3.4.1 Level A.

3.4.1.1 Case, Receiver-Transmitter CY-944/TPX or CY-944A/TPX with contents. Each case and contents, packaged as specified in 3.3, shall be individually packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, variety DW. Closure shall be in accordance with the appendix of the box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV or PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the box specification.

3.4.1.2 Case, Standardized Components, Electrical CY-951( )/TPX with contents. Each case and contents, packaged as specified in 3.3, shall be individually packed as specified in 3.4.1.1.

3.4.1.3 Cable Assembly CX-2287( )/U and CX-4443/U on reel. Each cable assembly or assemblies on reel, packaged as specified in 3.3, shall be individually packed as specified in 3.4.1.1.

3.4.1.4 Case, Antenna CY-1230( )/GPA-8 with contents. Each case with contents, packaged as specified in 3.3, shall be individually packed as specified in 3.4.1.1.

3.4.1.5 Cable Assemblies CX-4444( )/U for use with AN/TPX-26 only. The two cable assemblies, packaged as specified in 3.3, shall be packed as specified in 3.4.1.1.

3.4.1.6 Accessories, spares and ancillary items. Accessories, spares and ancillary items, packaged as specified in 3.3, shall be packed as specified in 3.4.1.1.

3.4.1.7 The items, packed as specified in 3.4.1.1 through 3.4.1.6, shall be arranged in the following sequence:

Item	AN/TPX-19	AN/TPX-20	AN/TPX-26	AN/TPX-27
CY-944/TPX	Box 1/4	Box 1/4		
CY-944A/TPX			Box 1/5	Box 1/4
CY-951( )/TPX	Box 2/4	Box 2/4	Box 2/5	Box 2/4
CX-2287( )/U	Box 3/4			
CX-2287( )/U and CX-4443/U			Box 3/5	
CY-1230( )/GPA-8		Box 3/4		Box 3/4
CX-4444( )/U			Box 4/5	
Accessories, Spares and Ancillary Items	Box 4/4	Box 4/4	Box 5/5	Box 4/4

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3.4.2 Level B. Radar Interrogator Set AN/TPX-19, 20, 26 or 27, packaged as specified in 3.3, shall be packed as specified in 3.4.1.

3.4.3 Level C. Each Radar Interrogator Set AN/TPX-19, 20, 26 and 27, packaged as specified in 3.3, shall be packed in shipping containers that comply with Federal or Military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage, during limited shipment, handling and intransit storage.

3.5 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

3.6 Package performance. Package testing shall be performed in accordance with MIL-P-116. The rough handling test will be performed only when invoked in the bid request or contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendments thereto.

b. Levels of packaging and packing (see section 3).

c. When the rough handling test is required.

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Preparing Activity:
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BILL OF MATERIAL
TABLE I

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
1	Case, Electrical CY-944/TPX containing:	1	III	25-7/16	14-1/8	12-7/16	For AN/TPX-19 and 20					
	Receiver-Transmitter, Radio, RT-211( )/TPX	1		21	11-1/8	9-7/8						
	Case, Electrical CY-944A/TPX containing:	1	III	29	14-1/8	12-7/16	For AN/TPX-26 and 27					
	Receiver-Transmitter, Radio RT-211( )/TPX and Simulator, Recognition Signal	1		21	11-1/8	9-7/8						
	SM-140( )/TPX	1		6-1/2	3-5/8	3-3/8						
				Secure equipment within the case, close case, and fasten latches.								
2	Pad, Corner	4		14-1/8	4-5/8	4-5/8	PPP-F-320	CF	WR	SW	W5c	
3	Pad, Top & Bottom	2		25-7/16	14-1/2	3/4	PPP-F-320	CF	WR	SW	W5c	
				For AN/TPX-19 and 20 only.								
					or							
				29	14-1/2	3/4	PPP-F-320	CF	WR	SW	W5c	
				For AN/TPX-26 and 27 only.								

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TABLE II

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
1	Case, Standardized Components, Electrical CY-951( )/TPX	1		17-1/8	13-1/16	12-7/16						
	containing: Coder-Control Interrogator Set KY-97( )/TPX	1		14-7/8	9-1/8	5-1/4						
	and Cable Assembly CX-1601( )/U	2		122 Package	per MIL-C-55442.							
2	Pad, Corner	4		13-3/8	3-3/8	3	PPP-F-320	CF	WR	SW	W5c	
3	Pad, Top and Bottom	2		16-3/8	13-7/8	1/2	PPP-F-320	CF	WR	SW	W5c	

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TABLE III

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
1	Cable Assembly CX-2287( )/U on reel	1	III	155 ft. 6	in. long		For AN/TPX-19 and 26 only.					
	and Cable Assembly CX-4443/U	1	III	155 ft. 6	in. long		For AN/TPX-26 only.					
2	Liner	1					PPP-F-320	CF	WR	SW	W5c	
3	Case CY-1230( )/ GPA-8	1	III	59-3/8	15-3/4	17-7/16						
	containing: Antenna AT-352( )/ UPA-22	1		111 Open dimensions.	15-1/2	12-1/8	For AN/TPX-20 and 27 only.					
4	Pad, Top & Bottom	2		59-11/16	16-1/16		PPP-F-320	CF	WR	SW	W5c	
5	Pad, End	2		15-3/4	17-7/16		PPP-F-320	CF	WR	SW	W5c	
6	Pad, Side	2		59-11/16	17-7/16		PPP-F-320	CF	WR	SW	W5c	

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BILL OF MATERIAL
TABLE IV

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
1	Control, Remote Switching C-1179( )/TPX-19	1	IC-1	5-3/4	2-1/8	8-1/4	For AN/TPX-19 and 26 only.					
	Paperboard, Cushioning	1		18	12		PPP-P-291	I				
	Tape	2		4	2		PPP-T-45	III			A	
	Bag	1		16	12		MIL-B-117	II	C			
	Bag, Cotton Duck CW-308( )/U	1	III	Fold nearly to proportionate dimensions.								
	Bag	1		12	8		MIL-B-117	II	C			
				Place folded cotton duck bag in paper bag, fold over opening and staple.								
	Cable Assembly CX-2288/U	1	Package per MIL-C-55442.				For AN/TPX-19 and 26 only.					
	Cable Assembly CX-2286/U	1	Package per MIL-C-55442.				For AN/TPX-19 and 20 only.					
	Cable Assembly Special Purpose	1	Package per MIL-C-55442.				For AN/TPX-19 and 26 only.					
	Cable Assembly CG-409( )/U	3	122 Package per MIL-C-55442.				For AN/TPX-19 only.					
		4	122 Package per MIL-C-55442.				For AN/TPX-26 only.					
		1	54 Package per MIL-C-55442.				For AN/TPX-19 and 26 only.					
		1	34 Package per MIL-C-55442.				For AN/TPX-20 and 27 only.					
		1	8 Package per MIL-C-55442.				For AN/TPX-26 and 27 only.					
	Cable Assembly CG-278( )/U	1	208 Package per MIL-C-55442.				For AN/TPX-19 and 26 only.					
		1	112 Package per MIL-C-55442.				For AN/TPX-20 and 27 only.					

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TABLE IV

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
	Cable Assembly CG-426( )/U	1		156								
			Package per MIL-C-55442.									
		1		120								
			Package per MIL-C-55442				For AN/TPX-26 only.					
		1		56								
			Package per MIL-C-55442				For AN/TPX-26 and 27 only.					
	Cable Assembly CG-4442( )/U	1		56								
			Package per MIL-C-55442.				For AN/TPX-26 and 27 only.					
	Cable Assembly CG-1060( )/U	1		102								
			Package per MIL-C-55442.				For AN/TPX-20 and 27 only.					
	Cable Assembly CX-2333( )/U	1		102								
			Package per MIL-C-55442.				For AN/TPX-20 and 27 only.					
	Cable Assembly CX-4443( )/U	1		210								
			Package per MIL-C-55442.				For AN/TPX-26 only.					
	Adapter UG-201( )/U	2	IC-1	1-5/16	13/16 dia.		For AN/TPX-19 and 26 only.					
	Cushioning, Cellulosic	2		2	2	1/4	PPP-C-843	II	B			
	Bag	2		3	3		MIL-B-117	II	C			
	Adapter, CG-212( )/U	1	IC-1				For AN/TPX-19 only.					
		2					For AN/TPX-26 only.					
	Cushioning, Cellulosic	3		2	2	1/4	PPP-C-843	II	B			
	Bag	3		3	3		MIL-B-117	II	C			
	Adapter, UG-274( )/U	1	IC-1				For AN/TPX-26 and 27 only.					
	Cushioning, Celluloisic	1		2	2	1/4	PPP-C-843	II	B			
	Bag	1		3	3		MIL-B-117	II	C			

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TABLE IV

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
	Interconnecting Box J-981( )/U	1	IC-1				For AN/TPX-26 only.					
	Twine	3		9	8 ply		T-T-871					
				Wind cable assembly into a coil of proportionate dimensions and tie in three places, securing the connectors in the ties within the eye of the coil.								
	Paperboard, Cushioning	1		30	9-1/4		PPP-P-291	I				
	Tape	2		4	2		PPP-T-45	III				
	Bag	1		13	13		MIL-B-117	II	C			
2	Relay Armature 1112-3-	1	IC-1									
	Cushioning, Cellulosic	1		2	2	1/4	PPP-C-843	II	B			
	Bag	1		3	3		MIL-B-117	II	C			
	Relay, Armature, Special	1	IC-1									
	Cushioning, Cellulosic	1		5	4	1/4	PPP-C-843	II	B			
	Bag	1		5	4		MIL-B-117	II	C			
	Relay, Thermal	1	IC-1									
	Paperboard, Cushioning	1		4	2-5/8		PPP-P-291	I				
	Bag	1		4	3		MIL-B-117	II	C			
	Fuses, Cartridge	25	III									
	Box, Paperboard	1		3	2	2	MIL-B-43014	A	a		Form I	II
	Lamp LM-52	2	III									
	Paperboard, Cushioning	1		3	3		PPP-P-291	I				
	Tape	1		3	2		PPP-T-45	III			A	
	Electron Tubes	23					Package per MIL-E-75.					
	Semi-Conductor Device	4					Package per MIL-S-19491.					
	Circuit Label	1	III									

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TABLE IV

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
	Paperboard, Cushioning	1		22	16-1/2		PPP-P-291	I				
	Tape	2		11-1/2	2		PPP-T-45	III				
3	Box, Fiberboard	1		12	6-1/2	5-1/2	PPP-B-636	CF	WR	SW	W6c	
				Place all parts listed under item 2 within the fiberboard box.								
4	Box, Fiberboard	1		19	16	11-1/2	PPP-B-636	CF	WR	SW	W5c	
				Place all parts listed under item 1 and the fiberboard box (item 3) within the fiberboard box (item 4).								

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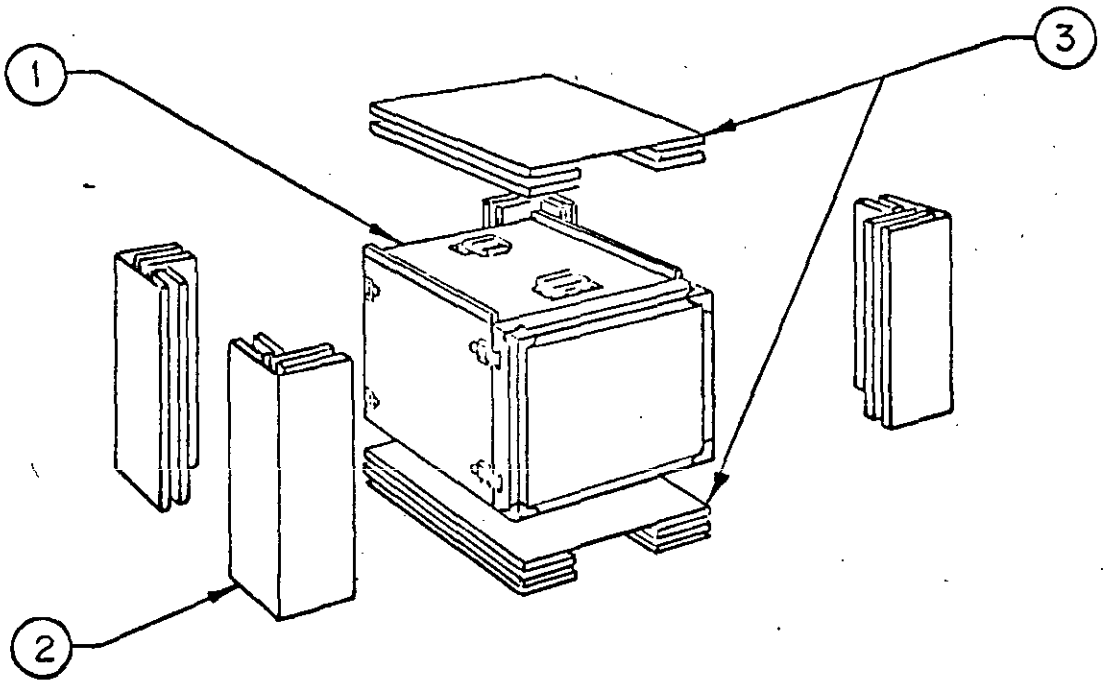


FIGURE 1

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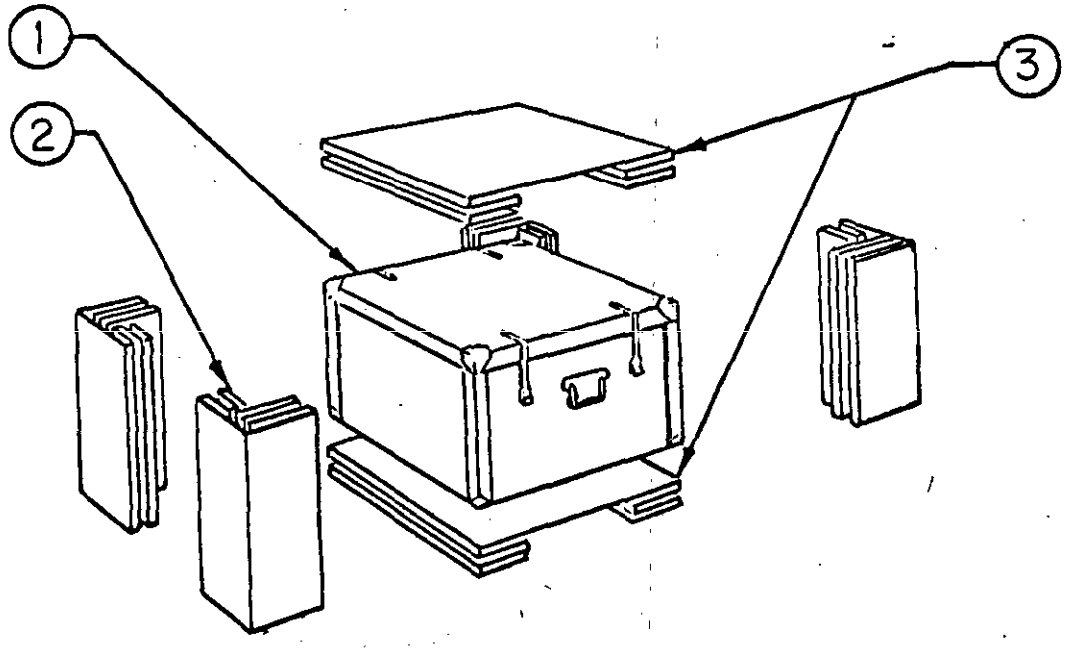


FIGURE 2

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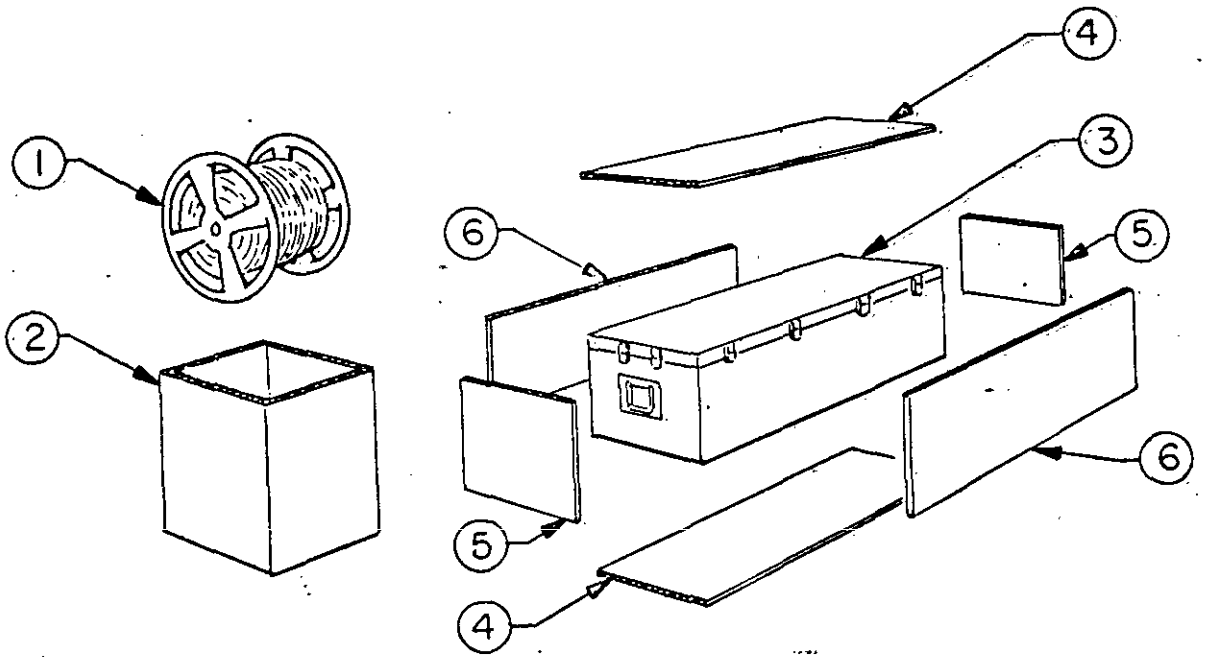


FIGURE 3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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